Transfer learning, Object Localization and Object Detection

Object Detection and Basic Object Detection Algorithms

Some of the Building Bocks for Object Detection

Courtesy: Andrew NG

What is Object localization and Detection?

- Object Detection
- Object Localization

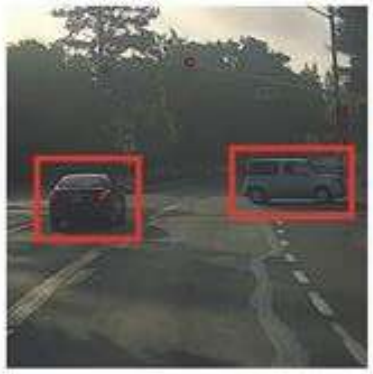
Image classification



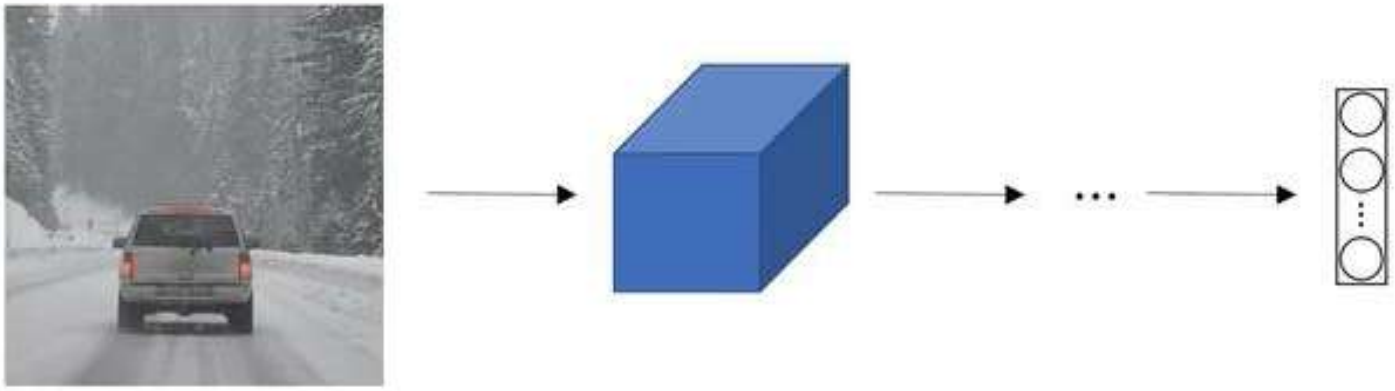
Classification with localization



Detection



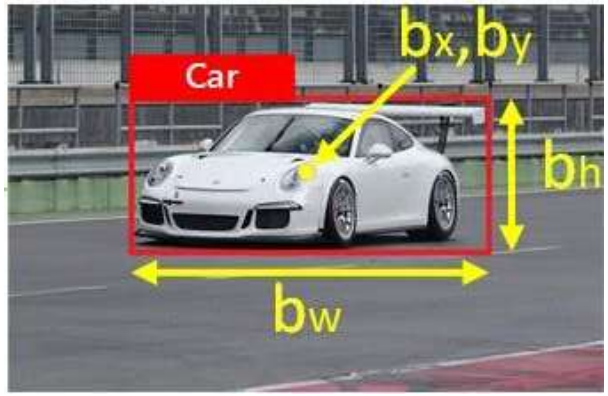
Classification with Localization



- Object categories for self-driving car
  1. Car 2. Bike, 3. Truck, 4.Background

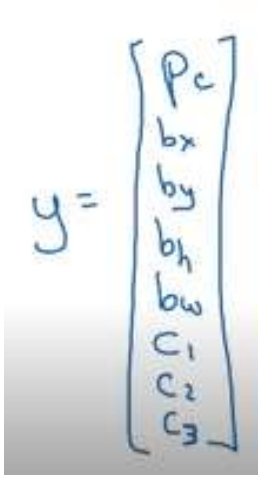
- Softmax with 4 possible outputs and

4 output units for bounding box (bx,by, bh, bw)

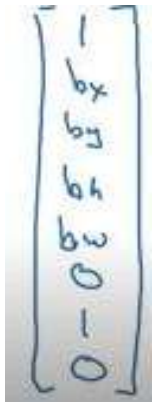


How to define the target label Y

- 4 Object categories: 1. Car 2. Bike, 3. Truck, 4.Background
- Need to output bx, by, bh, bw, and class labels (1-4)

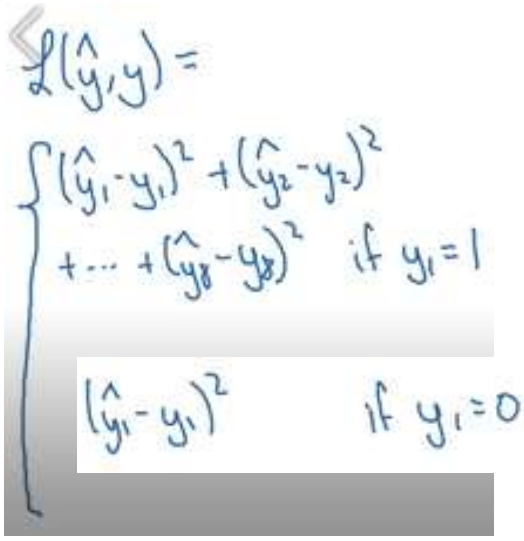












Landmark Detection

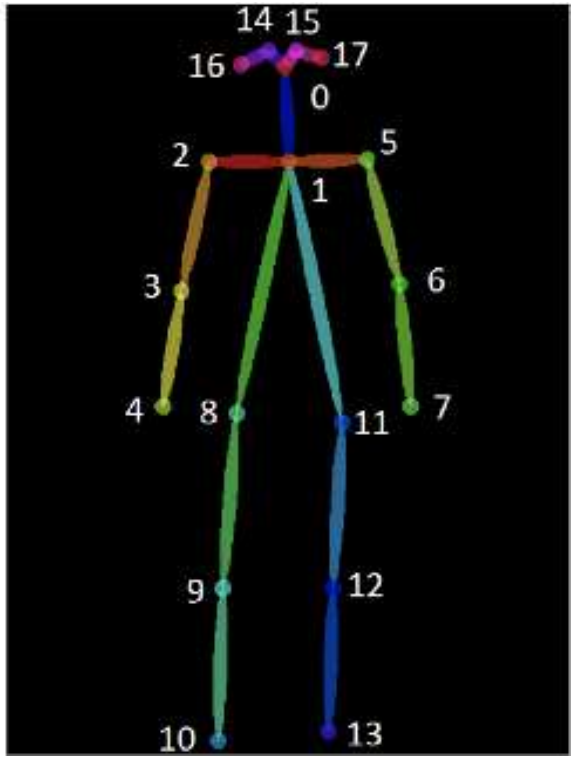


b_x, b_y, b_h, b_w



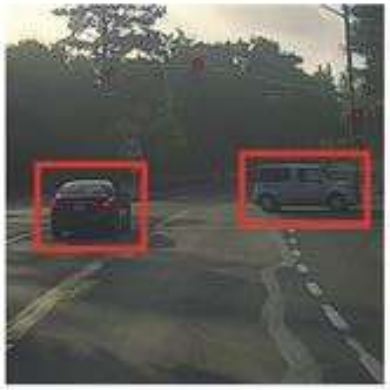
$l_{1x}, l_{1y},$
$l_{2x}, l_{2y},$
$l_{3x}, l_{3y},$
l_{4x}, l_{4y}
$\vdots$
l_{64}, l_{64y}





Object Detection

- Can we use Convolution Neural Net for Object Detection?
- Object Detection algorithm: Conv. Neural Net with sliding windows
- Car detection example

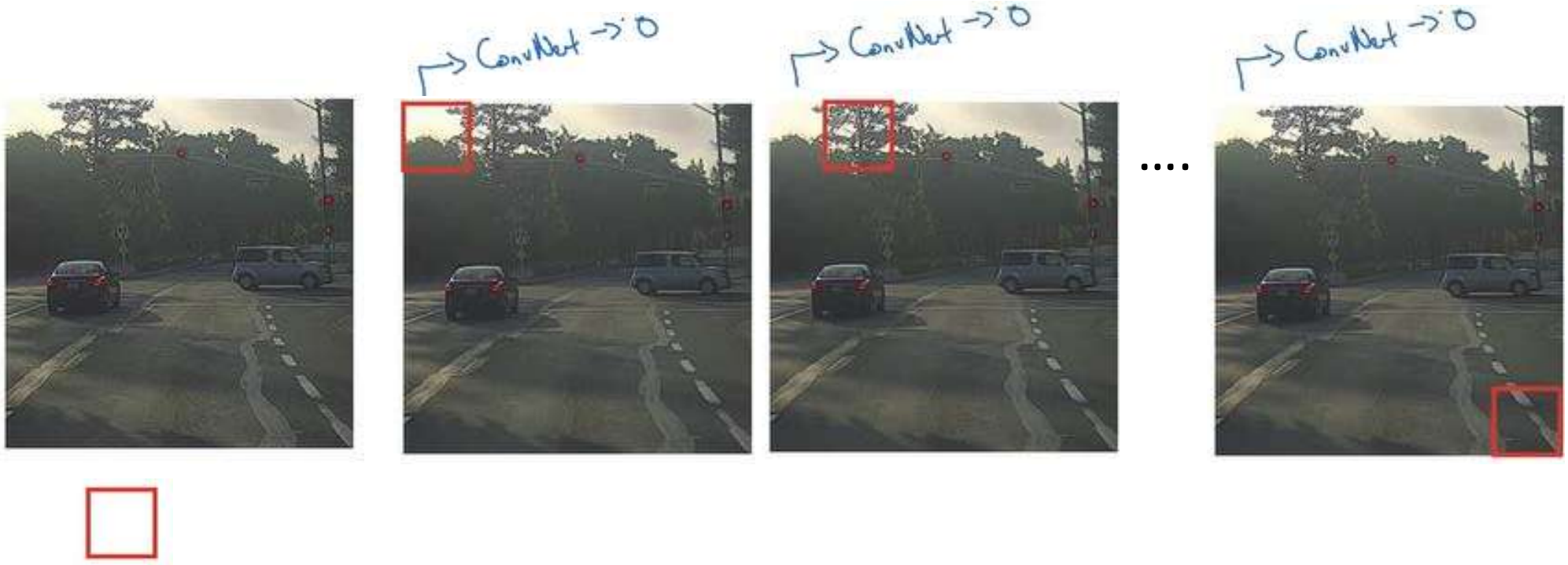






Sliding windows Detection (for Object detection)

- The trained ConvNet can be used in sliding windows detection. (Slide window across the entire image)



Sliding windows Detection (for Object detection)

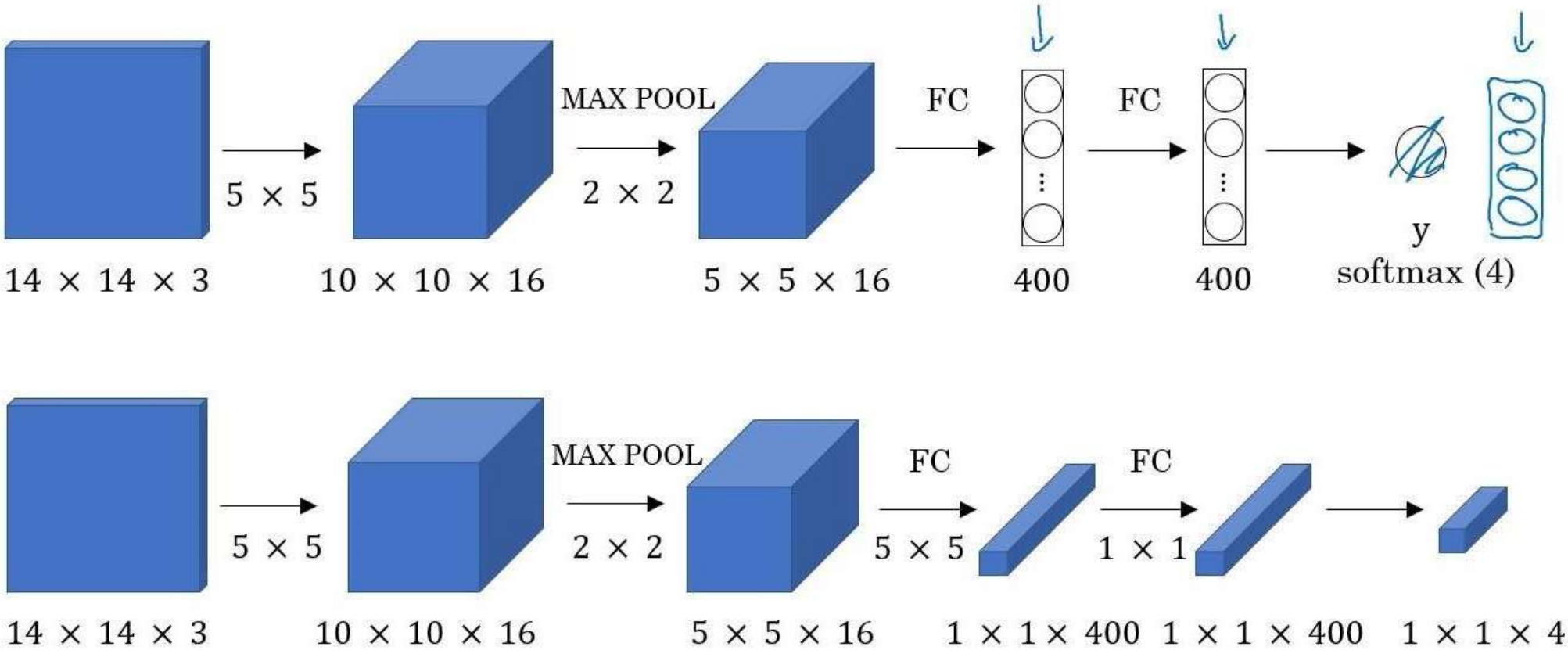
- Sliding window with Larger size



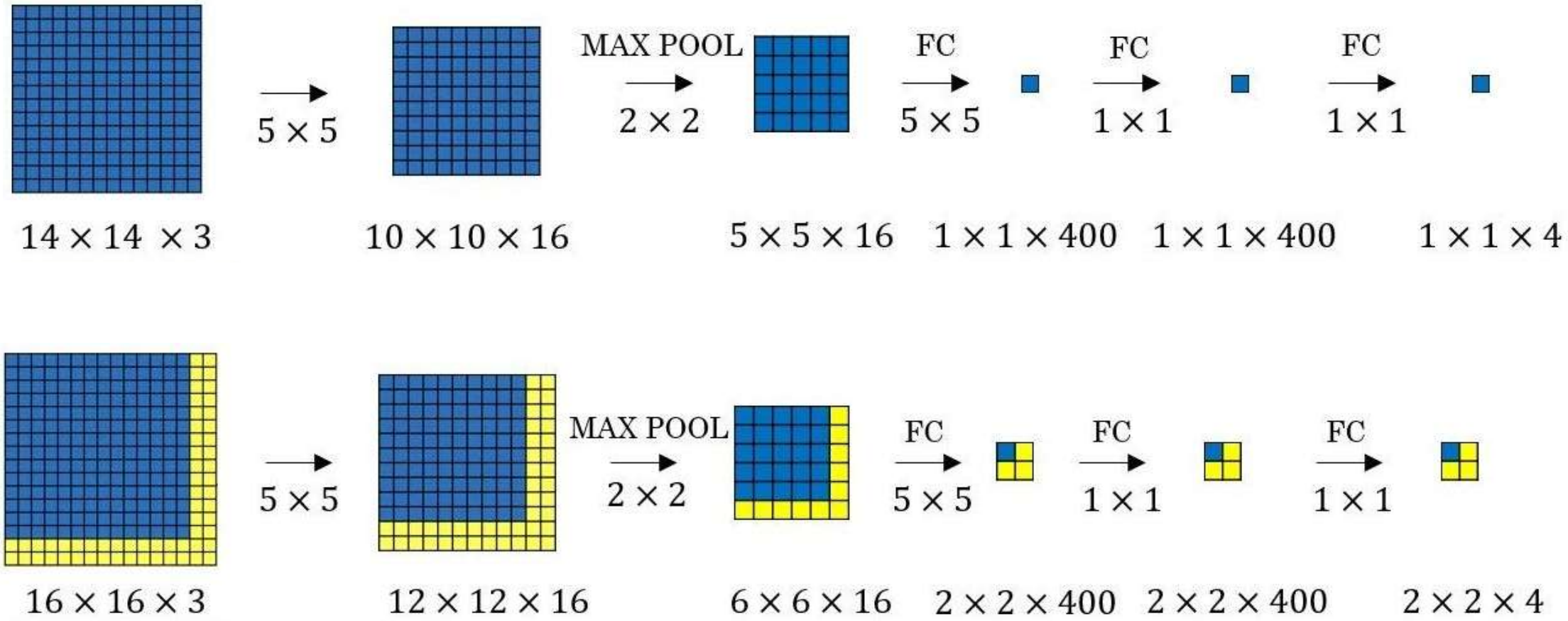
Convolution Implementation of Sliding Windows

Convolution Implementation of Sliding Windows

- Fully connected layers are concerted in convolution layers

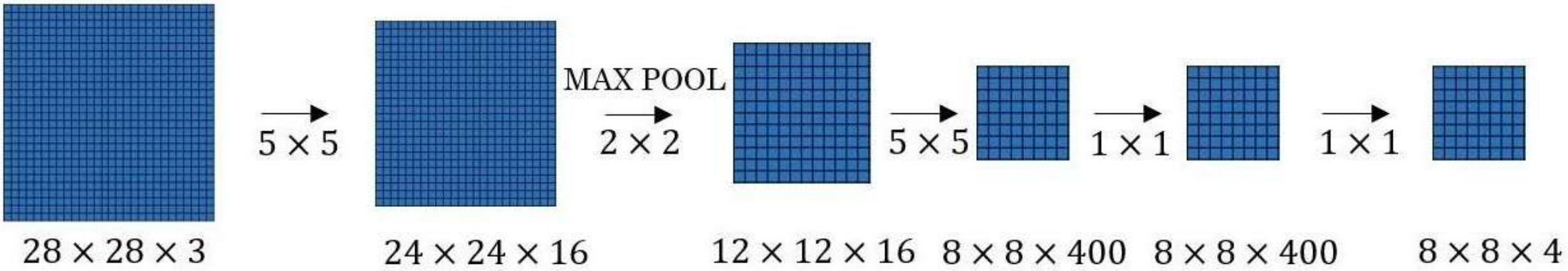


Convolution Implementation of Sliding Windows

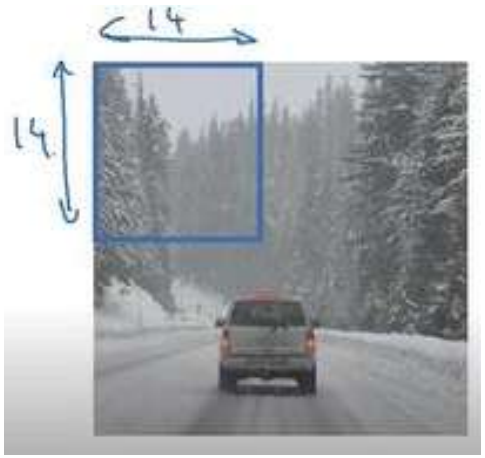


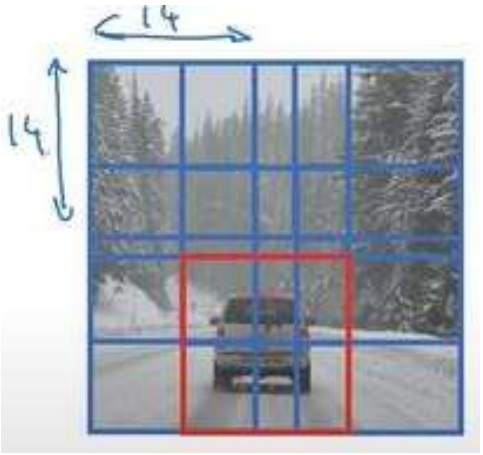
[Sermanet et al., 2014, OverFeat: Integrated recognition, localization and detection using convolutional networks]

Convolution Implementation of Sliding Windows





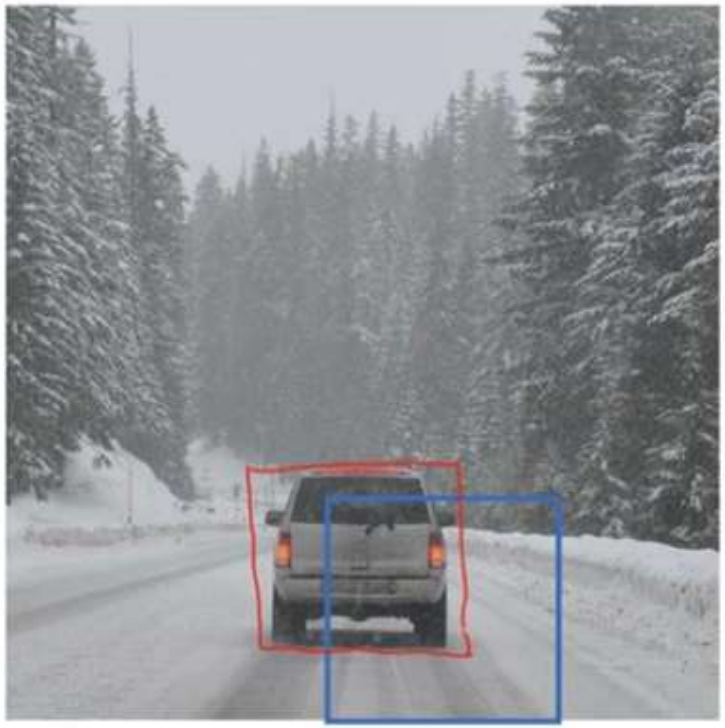




[Sermanet et al., 2014, OverFeat: Integrated recognition, localization and detection using convolutional networks]

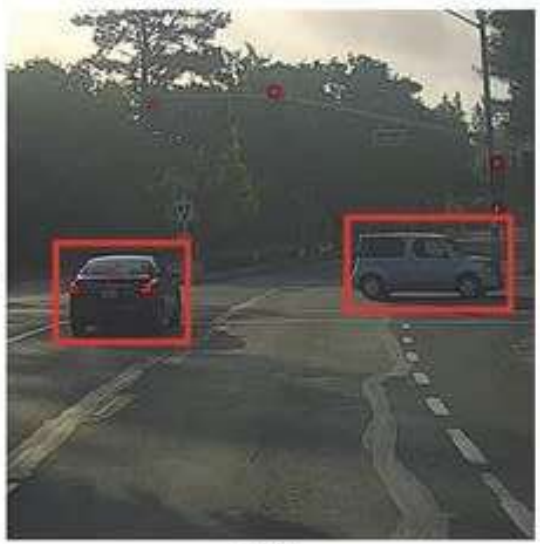
Convolution Implementation of Sliding Windows

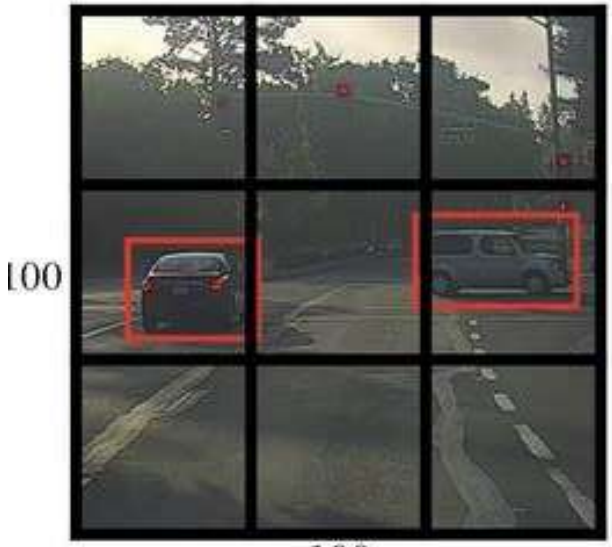
- Weakness:
  - Bounding boxes may not be really accurate



Accurate Object Detection (Bounding box prediction) using YOLO Algorithm

YOLO Algorithm





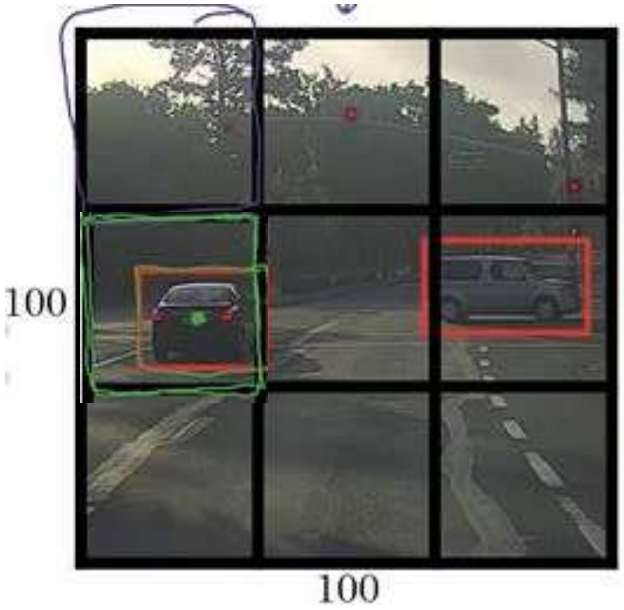
Labels for training

For each grid cell:

$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

YOLO Algorithm



$$\begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix} \qquad \begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Target output:

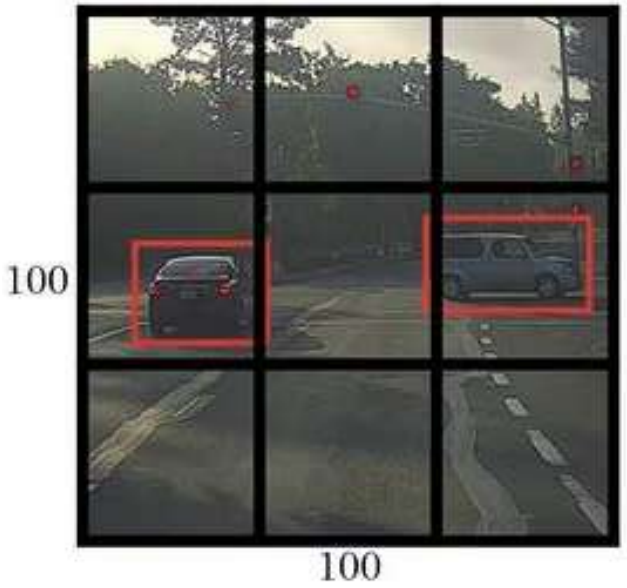
$3 \times 3 \times 8$

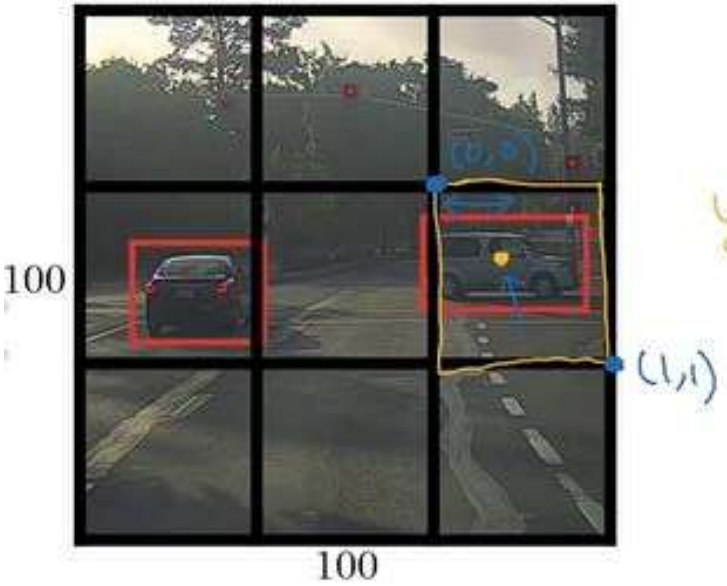


x → $100 \times 100 \times 3$ → CNN → MaxPool → .... → $3 \times 3 \times 8$ = y

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

How to specify the Bounding box?





$$y = \begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

0.4
0.3
0.9
0.5

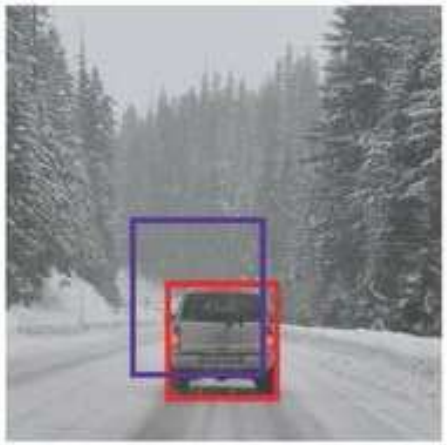
between 0 and 1

could be > 1

Evaluating Object Localization

Intersection Over Union (IoU): **It is a ratio between the intersection and the union of the predicted boxes and the ground truth boxes.** This stat is also known as the Jaccard Index and was first published by Paul Jaccard in the early 1900s.





$$IoU = \frac{\text{Intersection}}{\text{Union}}$$

Often used: correct if IoU >= 0.5

More generally, IoU is a measure of the overlap between two bounding boxes.

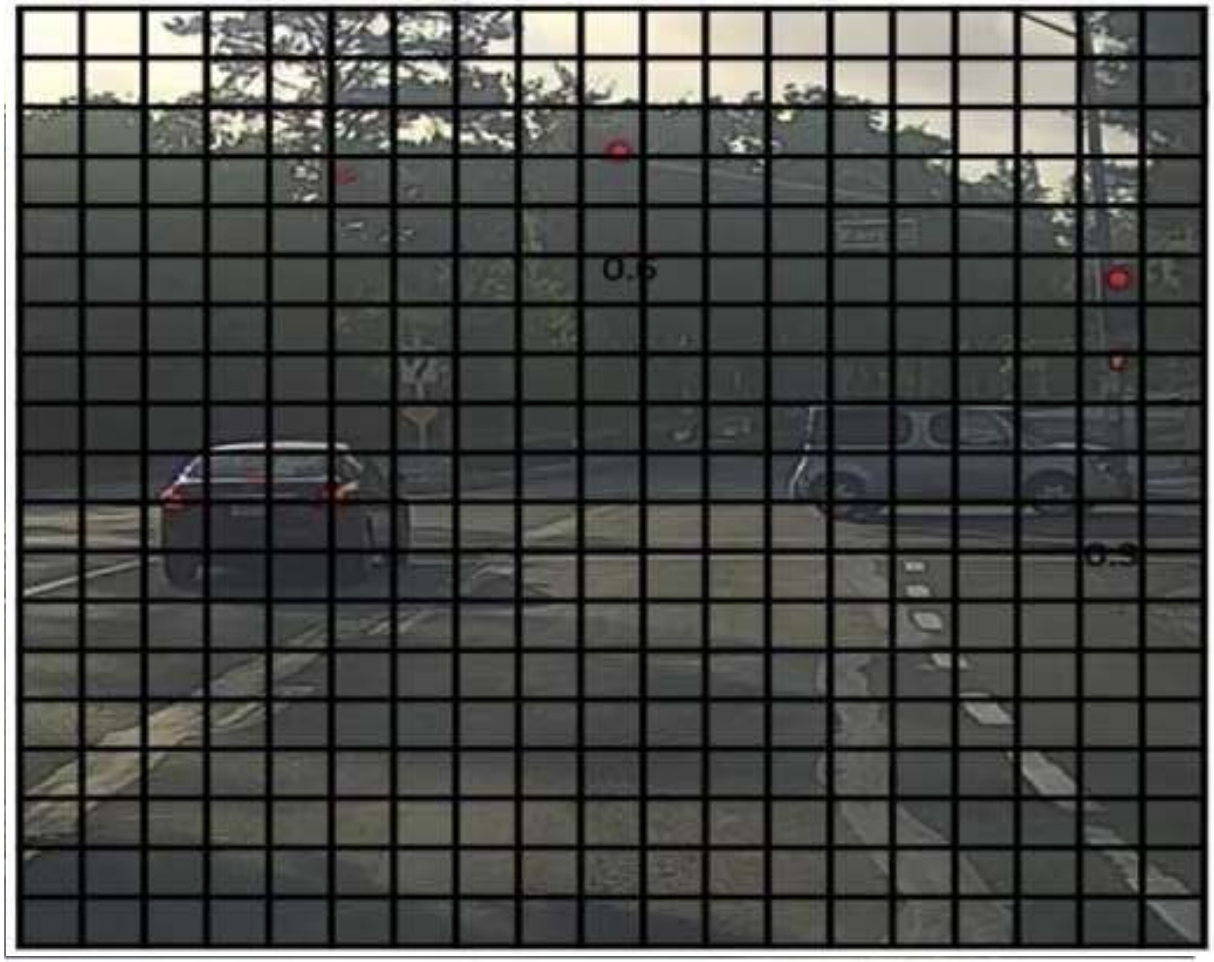
Object Detection:

Non-max Suppression

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

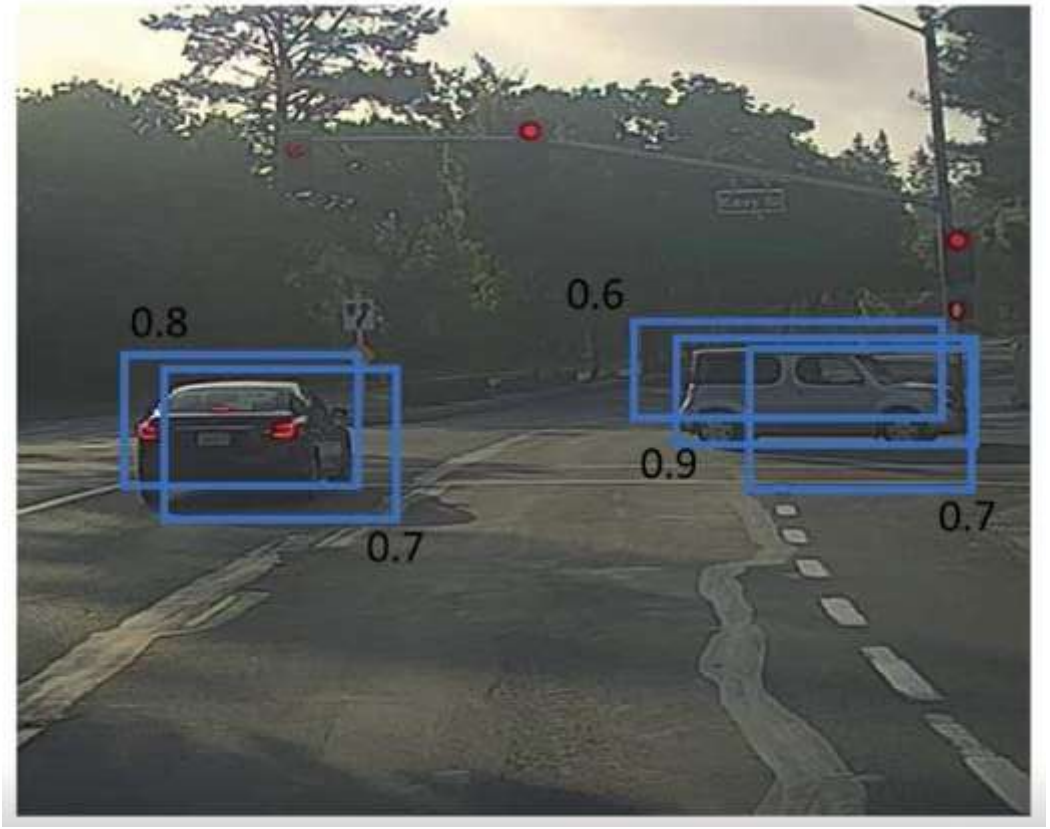
Non-max suppression

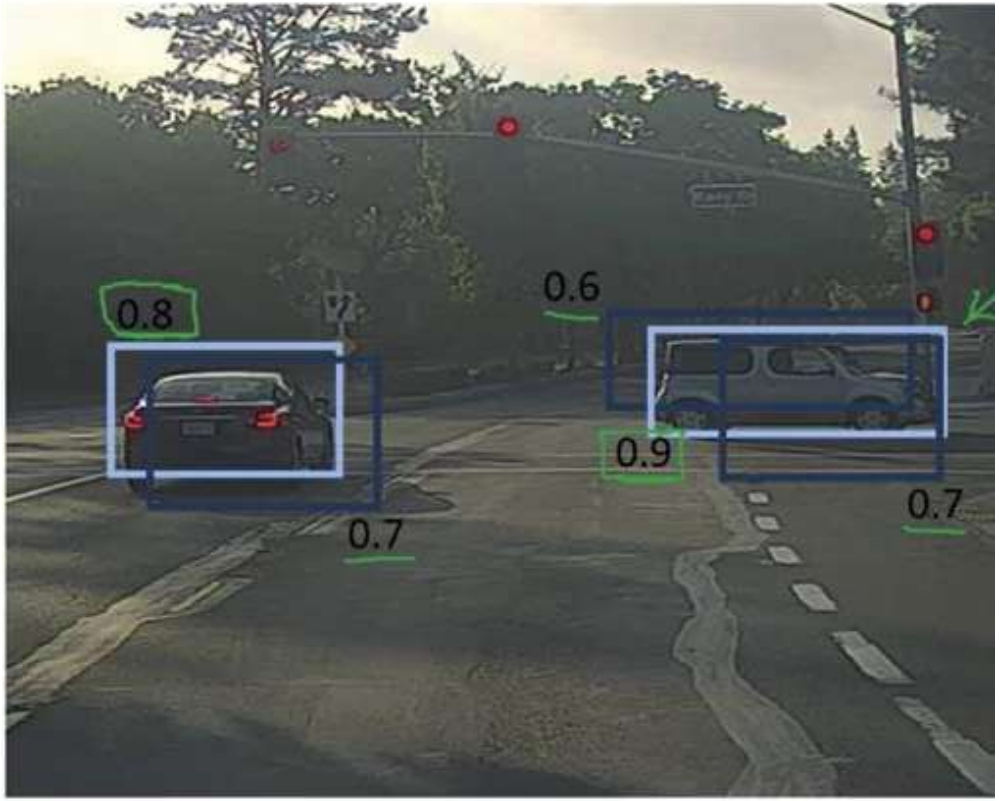
Example:



Non-max suppression

Example:

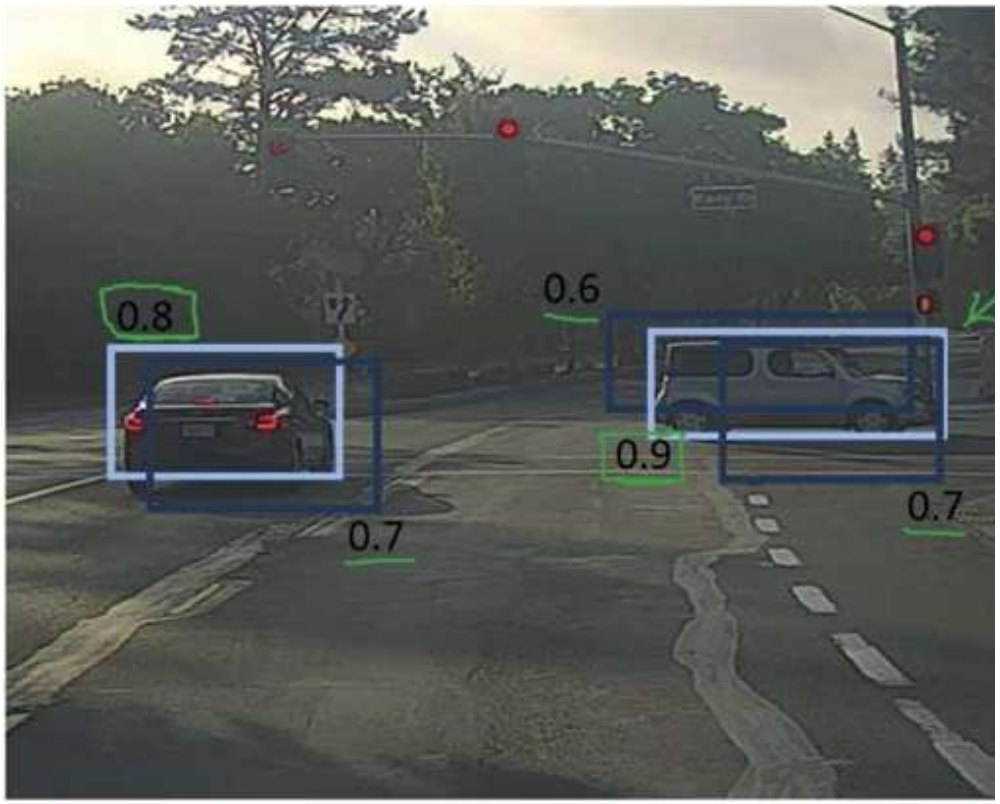


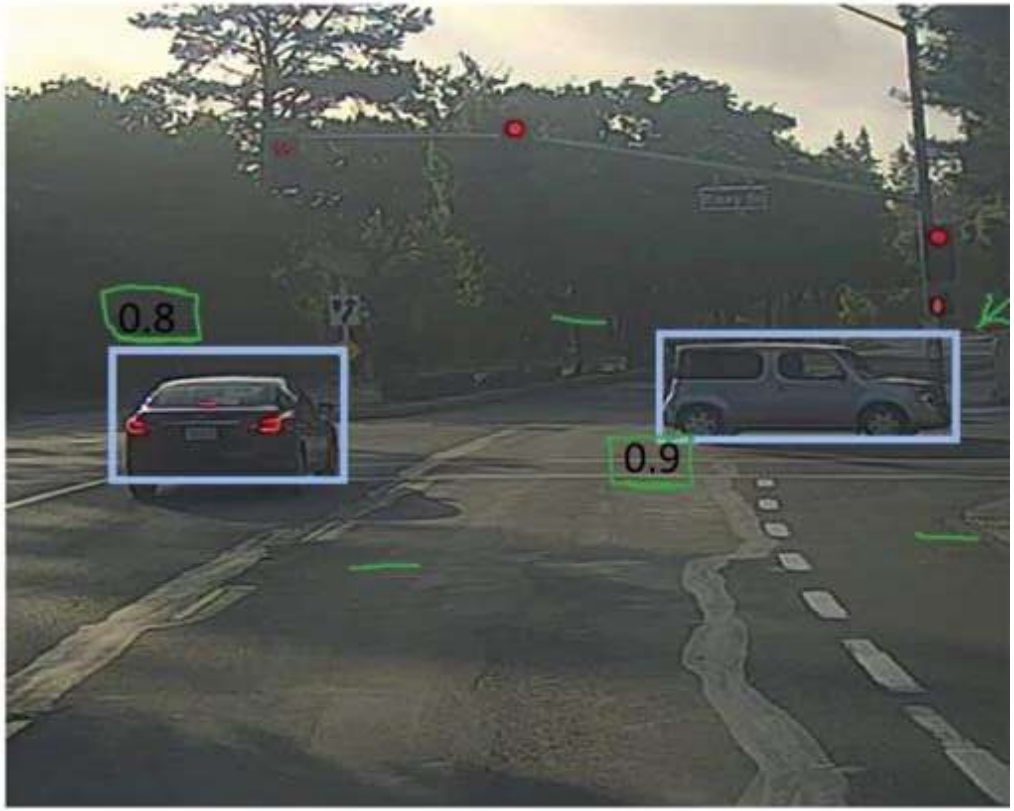


[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

Non-max suppression

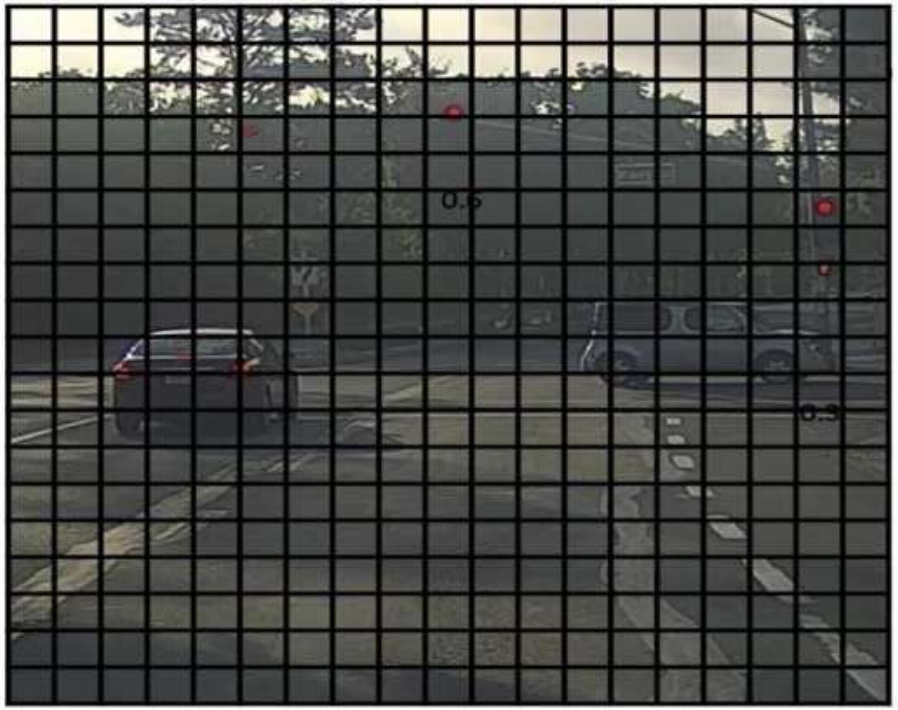
Example:





[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

Non-max suppression algorithm



Each output prediction is: $\begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \end{bmatrix}$

Discard all boxes with $p_c \leq 0.6$

While there are any remaining boxes:

- Pick the box with the largest p_c
  Output that as a prediction.
- Discard any remaining box with IoU ≥ 0.5 with the box output in the previous step

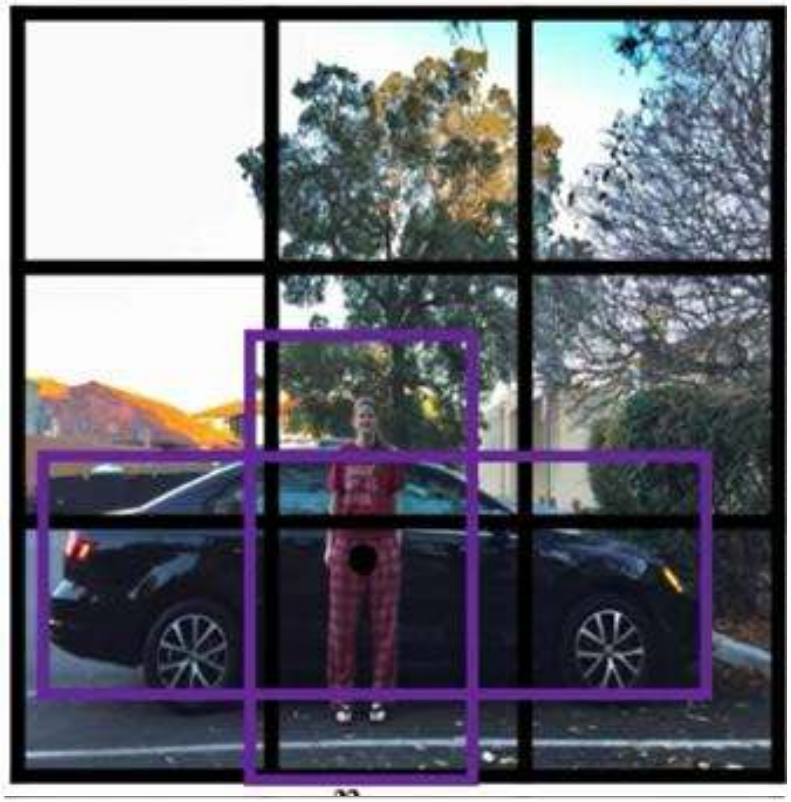
Object Detection:

Anchor Boxes

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

Anchor Boxes

Overlapping Objects



$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Anchor box 1:



Anchor box 2:



$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ \vdots \\ c_3 \end{bmatrix}$$

Anchor box 1

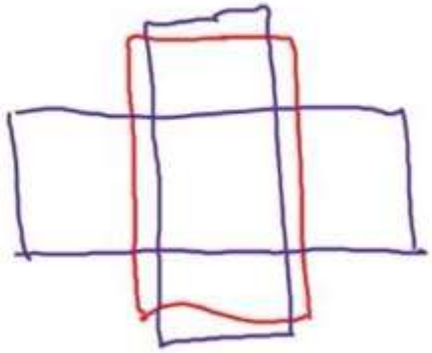
Anchor box 2

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

Anchor Box Algorithm

Previously:

Each object in training image is assigned to grid cell that contains that object's midpoint.



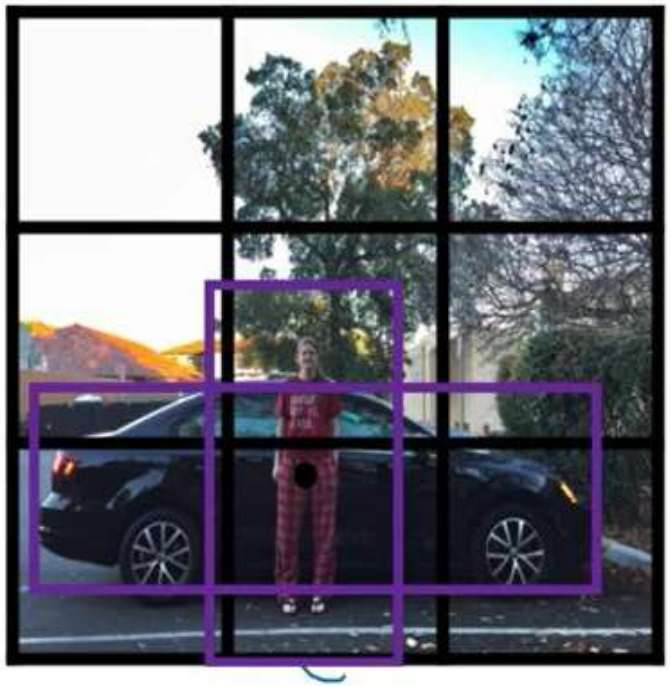
With two anchor boxes:

Each object in training image is assigned to grid cell that contains object's midpoint and anchor box for the grid cell with highest IoU.

(Grid cell, anchor box)

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

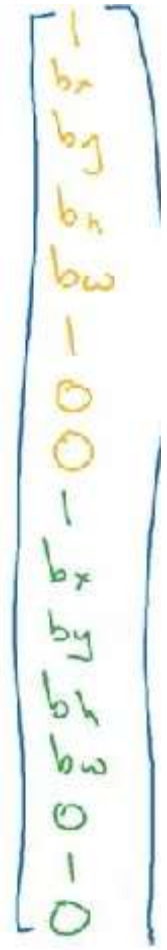
Anchor Box Example

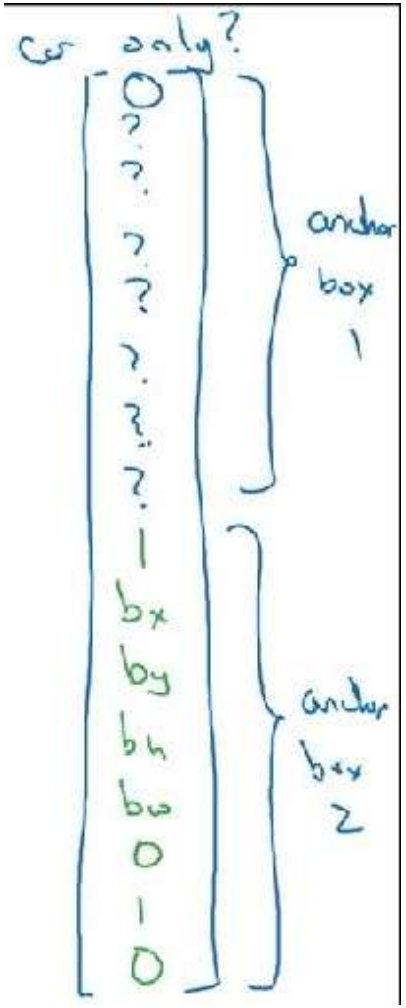


Anchor box 1: Anchor box 2:



$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$





[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

Overall:

YOLO Object Detection Algorithm

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

YOLO Algorithm

Training

1. Pedestrian
2. Car
3. Motorcycle

$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} \quad y = \begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix} \quad y = \begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

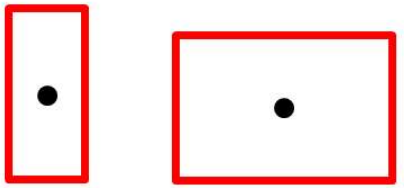


$y = 3 \times 3 \times 16$

$y = 3 \times 3 \times 2 \times 8$

anchors # parameters

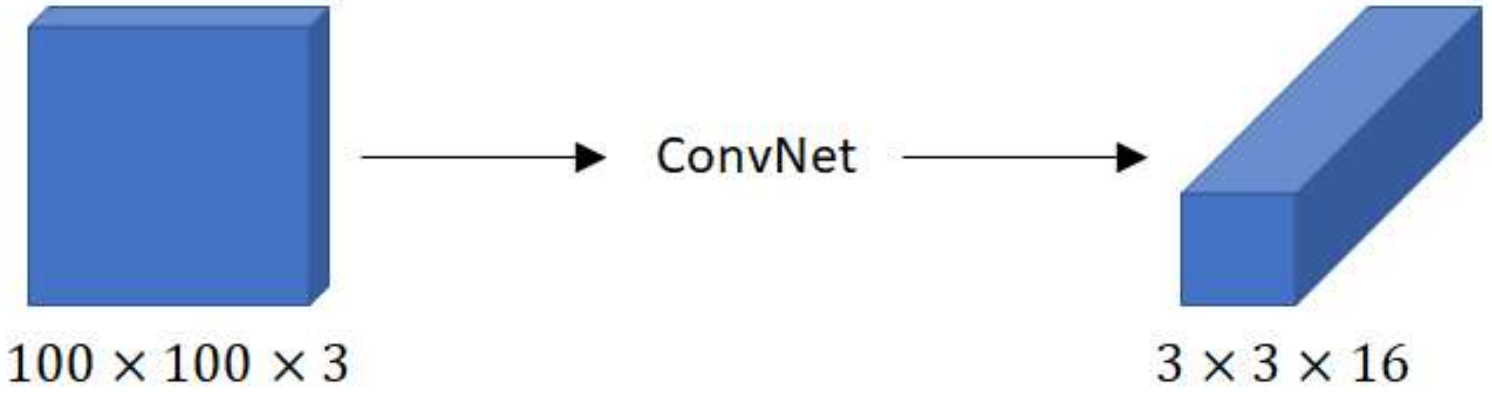
Anchor box 1: Anchor box 2:



[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

YOLO Algorithm

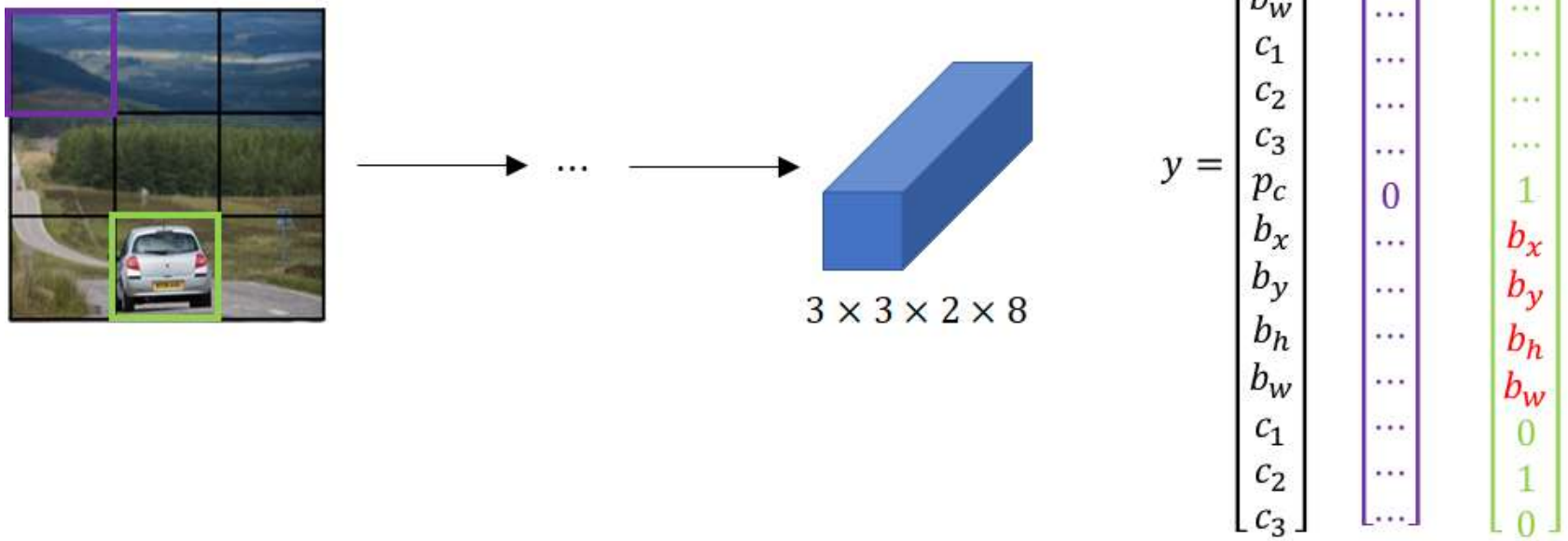
Training



[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

YOLO Algorithm

Prediction



[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]

YOLO Algorithm

Prediction: **How do we run a non-max suppression?**

- For each cell get 2 predicted bounding boxes
- Get rid of low probability predictions
- For each class (pedestrian, car, motorbike), use non-max suppression to generate final predictions.

[Redmon et al., 2015, You Only Look Once: Unified real-time object detection]